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Optical connector having fitted housings applying pressure to a protection sleeve

Abstract

An optical connector includes: a ferrule holding an optical fiber; a holder holding the ferrule; a protection sleeve including a tip that is fixed to a sleeve fixing portion of the holder; an inner housing in which at least a part of the sleeve fixing portion is housed; and an outer housing in which at least a part of the inner housing is housed. The inner housing includes a tab portion that tilts in a radial direction of the ferrule and applies a pressure on the protection sleeve by tilting inward in the radial direction in a state in which the tab portion is disposed inside the outer housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Japanese Patent Application No. 2021-039283, filed Mar. 11, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to an optical connector.

BACKGROUND

(3) Patent Document 1 discloses an optical connector having a configuration in which a built-in fiber held by a ferrule member is fusion-spliced to another optical fiber, and a fusion-spliced portion thereof is protected by a protection sleeve. The protection sleeve is fixed to a flange portion of the ferrule member by heat shrinkage.

PATENT LITERATURE

(4) [Patent Document 1]

(5) Japanese Unexamined Patent Application, First Publication No. 2011-95410

(6) Tension may act on the protection sleeve. As in Patent Document 1, when the protection sleeve is fixed only by a shrinkage force due to heating, a fixed state thereof tends to be unstable. As a result, there is a likelihood that the protection sleeve will fall off from the ferrule member due to the tension.

SUMMARY

(7) One or more embodiments of the present invention provide an optical connector in which a fixed state of a protection sleeve is made stable.

(8) An optical connector according to one or more embodiments of the present invention includes a ferrule holding an optical fiber, a holding part holding the ferrule, a protection sleeve whose tip is fixed to a sleeve fixing portion of the holding part, an inner housing inside which at least a part of the sleeve fixing portion is housed, and an outer housing inside which at least a part of the inner housing is housed, in which the inner housing includes a tab portion which is able to be tilted in a radial direction, and the tab portion is configured to apply a pressure on the protection sleeve by being tilted inward in the radial direction in a state in which the tab portion is positioned inside the outer housing.

(9) According to the embodiments described above, by positioning the tab portion inside the outer housing, it is possible to tilt the tab portion inward in the radial direction to apply a pressure on the protection sleeve. Due to the pressure, it is possible to fix the protection sleeve to the sleeve fixing portion. Therefore, it is possible to stabilize a fixed state, compared to a case in which the protection sleeve is fixed only by a structure whose fixed state is affected by heating such as, for example, a heat shrinkage force of the protection sleeve.

(10) Here, the tab portion may be disposed at a position facing the sleeve fixing portion.

(11) Also, the sleeve fixing portion may include a plurality of protrusions disposed at intervals in a longitudinal direction and protruding outward in the radial direction, and the tab portion may be disposed at a position facing at least a part of the plurality of protrusions.

(12) Also, an inner circumferential surface of the tab portion may have an arcuate shape when viewed from a longitudinal direction.

(13) Also, the tab portion may have a contact surface which comes in contact with the outer housing in a state in which a pressure is applied on the protection sleeve, and the contact surface may be inclined with respect to an inner surface of the outer housing.

(14) Also, a tensile strength member may be disposed inside of the protection sleeve, and the tensile strength member may be positioned between the tab portion and the sleeve fixing portion.

(15) According to the above-described embodiments of the present invention, it is possible to provide an optical connector in which a fixed state of the protection sleeve is made stable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view of an optical connector according to one or more embodiments.
(2) FIG. **2** is an exploded perspective view of the optical connector of FIG. **1**.
(3) FIG. **3** is a perspective view of an inner housing of FIG. **2**.
(4) FIG. **4** is a view in which a ferrule, a first member, and a second member of FIG. **2** are extracted.
(5) FIG. **5** is a cross-sectional view of the optical connector of one or more embodiments before a case and an outer housing are attached.
(6) FIG. **6** is a cross-sectional view taken along line VI-VI indicated by the arrow in FIG. **5**.
(7) FIG. **7** is a cross-sectional view of a state in which the outer housing is attached to the inner housing of FIG. **5**.

DETAILED DESCRIPTION

(8) Hereinafter, an optical connector of one or more embodiments will be described with reference to the drawings.
(9) As illustrated in FIGS. **1** and **2**, an optical connector **1** includes a case **10**, an outer housing **20**, an inner housing **30**, a ferrule **40**, a holding part (i.e., holder) **2**, a biasing member **60**, a protection unit **80**, and a boot **90**. The optical connector **1** is provided at an end portion of an optical cable C. The ferrule **40** includes a fiber hole **41**, and a connection end surface **42** at which the fiber hole **41** opens.
(10) (Definition of Directions)
(11) In one or more embodiments, a direction in which the fiber hole **41** extends is defined as a longitudinal direction X. One direction orthogonal to the longitudinal direction X is defined as a first direction Z, and a direction orthogonal to both the longitudinal direction X and the first direction Z is defined as a second direction Y. In the longitudinal direction X, the connection end surface **42** side of the ferrule **40** is defined as a front side (+X side) or a tip side. A side opposite to the front side is defined as a rear side (−X side) or a base end side. When viewed from the longitudinal direction X, a direction intersecting a central axis of the fiber hole **41** is defined as a radial direction, and a direction of revolving around the central axis is defined as a circumferential direction. The first direction Z may also be referred to as a vertical direction. One side in the first direction Z may be referred to as an upper side (+Z side). A side opposite to the upper side may be referred to as a lower side (−Z side). Also, one side in the second direction Y may be referred to as a left side (+Y side). A side opposite to the left side may be referred to as a right side (−Y side).
(12) As illustrated in FIG. **1**, the case **10** has a quadrangular tubular shape extending in the longitudinal direction X. The outer housing **20**, the inner housing **30**, the ferrule **40**, and the like are disposed inside of the case **10** in the radial direction. The ferrule **40** protrudes forward from the case **10**.
(13) As illustrated in FIG. **2**, the outer housing **20** has a quadrangular tubular shape extending in the longitudinal direction X. Note that, it is possible to change shapes of the case **10** and the outer housing **20** as appropriate and the shapes of the case **10** and the outer housing **20** may be, for example, cylindrical. A pair of locking holes **21** are formed on the outer housing **20**. The pair of locking holes **21** are formed on a right surface (a surface facing the right side) and a left surface (a surface facing the left side) of the outer housing **20**, respectively. That is, the pair of locking holes **21** are formed to be radially symmetrical with respect to the fiber hole **41**.
(14) As illustrated in FIG. **3**, the inner housing **30** includes a first tubular part **30***a* and a second

tubular part **30***b* positioned behind the first tubular part **30***a*. The first tubular part **30***a* has a tubular shape extending in the longitudinal direction X and has a quadrangular cross section. The second tubular part **30***b* has a cylindrical shape extending in the longitudinal direction X. At least a part of the first tubular part **30***a* is housed inside the outer housing **20**.

(15) A slit **31** and a tab portion **32** are formed on each of an upper surface (a surface facing upward) and a lower surface (a surface facing downward) of the first tubular part **30***a*. That is, the inner housing **30** includes a pair of slits **31** and a pair of tab portions **32** symmetrically in the radial direction. The upper and lower slits **31** and tab portions **32** have shapes that are vertically symmetrical to each other. Each of the slits **31** has a C shape that opens toward the rear when viewed from the first direction Z. A portion of the inner housing **30** surrounded by the slit **31** is the tab portion **32**. A base end portion (an end portion on the rear side) of the tab portion **32** is connected to the first tubular part **30***a*.

(16) The tab portion **32** is a portion that is elastically deformable with respect to the first tubular part **30***a*. The tab portion **32** is configured to be elastically deformable in the first direction Z with the base end portion (base end side) as a base point when a tip side of the tab portion **32** receives an external force. That is, in a case of one or more embodiments, the tab portion **32** positioned on the upper side is able to be tilted downward, and the tab portion **32** positioned on the lower side is able to be tilted upward. A protruding portion **36** that protrudes outward in the first direction Z is formed at a tip portion (an end portion on the front side) of the tab portion **32**. The protruding portion **36** has an inclined surface **36***a* (first inclined surface) and a contact surface **36***b*.

(17) The inclined surface **36***a* is provided to facilitate insertion of the inner housing **30** into the outer housing **20** from behind. The inclined surface **36***a* is inclined outward in the first direction Z toward the rear. The contact surface **36***b* is positioned behind the inclined surface **36***a*. The contact surface **36***b* comes in contact with an inner surface of the outer housing **20** when the inner housing **30** is inserted into the outer housing **20** (see FIG. **7**). Details of the contact surface **36***b* will be described later.

(18) As illustrated in FIG. **3**, a key groove **35** is formed on the upper surface of the first tubular part **30***a*. The key groove **35** is recessed rearward from a front end of the inner housing **30**.

(19) A locking piece **33** and two slits **34** are formed on each of a right surface (a surface facing the right side) and a left surface (a surface facing the left side) of the first tubular part **30***a*. The locking piece **33** and slits **34** on the left and right have shapes that are bilaterally symmetrical to each other in the radial direction. The two slits **34** formed on each of the right surface and the left surface of the first tubular part **30***a* are disposed with a distance therebetween in the first direction Z and extend in the longitudinal direction X. A portion disposed between the two slits **34** is the locking piece **33**.

(20) A locking protrusion **33***a* is formed at a central portion of the locking piece **33** in the longitudinal direction X. Each of the locking protrusions **33***a* protrudes outward in the second direction Y than the right surface or the left surface of the first tubular part **30***a*. The inner housing **30** is inserted into the outer housing **20** from a rear side of the outer housing **20**. Then, the locking protrusions **33***a* are locked in the locking holes **21** of the outer housing **20**, and thereby the inner housing **30** is connected to the outer housing **20**. In order to facilitate connection of the inner housing **30** to the outer housing **20**, an inclined surface **33***b* (second inclined surface) that inclines outward in the second direction Y toward the rear is formed on the locking protrusion **33***a*.

(21) As illustrated in FIG. **4**, the ferrule **40** has a cylindrical shape extending in the longitudinal direction X. As a material of the ferrule **40**, for example, it is possible to employ zirconia. An optical fiber F is inserted into the fiber hole **41** of the ferrule **40** (see FIG. **5**). The optical fiber F is fixed to the ferrule **40** by, for example, an adhesive. An end portion of the optical fiber F is exposed at the connection end surface **42**.

(22) As illustrated in FIG. **4**, a rear end portion of the ferrule **40** is held by the holding part **2**. The holding part **2** of one or more embodiments has a structure in which a first member **50** and a second

member **70**, which are separate bodies, are combined. However, the holding part **2** may be a single member (for example, a member in which the first member **50** and the second member **70** are integrated).

(23) The first member **50** includes a flange portion **51** and a sliding tubular portion **52**. The flange portion **51** has a tubular shape and surrounds a rear end portion of the ferrule **40**. The ferrule **40** is fitted inside the flange portion **51**. A rear end surface **51***a* of the flange portion **51** is biased forward by the biasing member **60** (see FIG. **2**). Therefore, the ferrule **40** held by the flange portion **51** also receives a forward biasing force.

(24) As illustrated in FIG. **4**, the sliding tubular portion **52** extends rearward from the flange portion **51**. An outer diameter of the sliding tubular portion **52** is smaller than an outer diameter of the flange portion **51**. The sliding tubular portion **52** is positioned inside a surrounding tubular portion **71** of the second member **70**. The sliding tubular portion **52** is slidable in the longitudinal direction X with respect to the surrounding tubular portion **71**. A slide rib **53** protruding upward is formed on the sliding tubular portion **52**. The slide rib **53** extends linearly in the longitudinal direction X. The slide rib **53** is positioned inside a slide groove **72** formed in the surrounding tubular portion **71** and slides with respect to the slide groove **72**. A pair of retainers **54** are formed at a rear end portion of the sliding tubular portion **52**. The retainers **54** protrude outward in the second direction Y from the sliding tubular portion **52**.

(25) The retainers **54** are positioned inside a pair of holes **75** formed in the surrounding tubular portion **71**, respectively. Thereby, the first member **50** and the second member **70** are connected. As illustrated in FIG. **4**, a dimension of the retainer **54** is smaller than a dimension of the hole **75** in the longitudinal direction X. In a state in which the retainer **54** is positioned at a front end portion of the hole **75**, a gap in the longitudinal direction X is provided between a front end surface **71***a* of the surrounding tubular portion **71** and the rear end surface **51***a* of the flange portion **51**. The first member **50** is able to slide rearward with respect to the second member **70** within a range of the gap.

(26) The second member **70** includes a surrounding tubular portion **71**, a sleeve fixing portion **73**, and a second flange portion **74**. The surrounding tubular portion **71** has a cylindrical shape extending in the longitudinal direction X. The second flange portion **74** is positioned at a rear end portion of the surrounding tubular portion **71** and protrudes outward in the radial direction. The second flange portion **74** has a spring receiving surface **74***a* facing forward. When the biasing member **60** is compressed between the rear end surface **51***a* of the flange portion **51** and the spring receiving surface **74***a* of the second flange portion **74**, the biasing force described above is generated (see FIG. **2**). As the biasing member **60**, for example, it is possible to use a coil spring. A key portion **76** protruding upward is formed on the second flange portion **74**. Although not illustrated, the key portion **76** is inserted into the key groove **35** of the inner housing **30** in a state in which the optical connector **1** has been assembled.

(27) The sleeve fixing portion **73** has a cylindrical shape extending rearward from the second flange portion **74**. An outer diameter of the sleeve fixing portion **73** is smaller than an outer diameter of the surrounding tubular portion **71**. A plurality of protrusions **73***a* are formed on the sleeve fixing portion **73**. The plurality of protrusions **73***a* protrude outward in the radial direction from the sleeve fixing portion **73** and are disposed at intervals in the longitudinal direction. A front end portion of the protection unit **80** is fixed to the sleeve fixing portion **73** (see FIG. **5**).

(28) When the optical connector **1** is connected to another optical connector, the connection end surface **42** of the ferrule **40** abuts against a ferrule of another optical connector **1**. At this time, the ferrule **40** retreats, against the forward biasing force due to the biasing member **60**. More specifically, the ferrule **40** and the first member **50** integrally retreat with respect to the second member **70** while compressing the biasing member **60** in the longitudinal direction X. At this time, the optical fiber F is able to be bent inside the surrounding tubular portion **71**.

(29) FIG. **5** is a cross-sectional view of the optical connector **1** of one or more embodiments before

the case **10** and the outer housing **20** are attached. FIG. **6** is a cross-sectional view taken along line VI-VI indicated by the arrow in FIG. **5**. FIG. **7** is a cross-sectional view of a state in which the outer housing **20** is attached to the inner housing **30** of FIG. **5**. Note that, FIGS. **2** and **6** illustrate a state before a protection sleeve **81** is heat-shrunk. FIGS. **5** and **7** illustrate a state after the protection sleeve **81** is heat-shrunk.
(30) As illustrated in FIGS. **5** and **6**, the protection unit **80** includes the protection sleeve **81**, a pair of tensile strength members **82**, a rod-shaped tensile strength member **83**, and an adhesive layer **84**. The protection sleeve **81** has a tubular shape extending in the longitudinal direction X. As the protection sleeve **81**, it is possible to use a heat-shrinkable tube.
(31) The protection sleeve **81** before being fixed to the sleeve fixing portion **73** has an inner diameter larger than a maximum outer diameter of the plurality of protrusions **73***a* formed on the sleeve fixing portion **73**. When the protection sleeve **81** is heated with the protection sleeve **81** covering the sleeve fixing portion **73** and the plurality of protrusions **73***a*, the protection sleeve **81** is heat-shrunk. Thereby, as illustrated in FIGS. **5** and **7**, the protection sleeve **81** shrinks to fasten the protrusions **73***a*, and the protection sleeve **81** is fixed to the sleeve fixing portion **73**.
(32) The pair of tensile strength members **82** and the rod-shaped tensile strength member **83** are disposed inside of the protection sleeve **81**. As illustrated in FIG. **5**, the tensile strength members **82** have a sheet shape extending in the longitudinal direction X. As the tensile strength member **82**, for example, it is possible to use a cloth obtained by weaving fibers or a sheet material obtained by hardening fibers with a resin. As the fibers, for example, it is possible to use aramid fibers, glass fibers, or the like. With the protection sleeve **81** fixed to the sleeve fixing portion **73**, each of the tensile strength members **82** is sandwiched between the protrusions **73***a* and the protection sleeve **81**. The pair of tensile strength members **82** are disposed on an upper side and a lower side of the sleeve fixing portion **73**, respectively.
(33) The rod-shaped tensile strength member **83** has a rod-shape (for example, a columnar shape) extending in the longitudinal direction X. It is possible to use any material as the rod-shaped tensile strength member **83**. The rod-shaped tensile strength member **83** is positioned behind the sleeve fixing portion **73**. The adhesive layer **84** is formed of a material that is softened by heating such as, for example, a hot-melt adhesive. In one or more embodiments, as illustrated in FIG. **6**, the adhesive layer **84** in a state before assembly of the optical connector **1** has a tubular shape and is disposed on a radially inner side of the pair of tensile strength members **82** and the rod-shaped tensile strength member **83**. When the protection sleeve **81** is heated, the adhesive layer **84** is softened, and at least a part thereof adheres to and integrates with an inner circumferential surface of the protection sleeve **81**. In FIGS. **5** and **7**, illustration of the adhesive layer **84** is omitted. The adhesive layer **84** may be applied, for example, on the inner circumferential surface of the protection sleeve **81** in advance.
(34) As illustrated in FIG. **5**, the optical fiber F has a structure in which two optical fibers (the built-in fiber F**1** and the splicing fiber F**2**), which have been originally separated, are fusion-spliced at a fusion-splicing point P. That is, the optical connector **1** of one or more embodiments is a so-called fusion-splicing connector. The protection unit **80** has a role of covering and protecting the fusion-splicing point P. A built-in fiber F**1** is an optical fiber fixed to the ferrule **40** in advance before the built-in fiber F**1** and a connection fiber F**2** are fusion-spliced. The connection fiber F**2** is an optical fiber that has been built in the optical cable C (see FIG. **1**).
(35) Work of providing the optical connector **1** to an end portion of the optical cable C, that is, work of fusion-splicing the built-in fiber F**1** and the connection fiber F**2** may be performed at a site in which the optical cable C is laid. Therefore, work of heating the protection sleeve **81**, that is performed after the fusion splicing, may also be performed at the site in which the optical cable C is laid.
(36) Note that, a plurality of optical connectors **1** may be provided for one optical cable C.
(37) Next, an assembling method (manufacturing method) of the optical connector **1** will be

described.

(38) First, the ferrule **40** is inserted and fixed inside the flange portion **51** of the first member **50**. Thereafter, the built-in fiber F**1** is inserted through the ferrule **40** via the first member **50** and is fixed by an adhesive with an end portion of the built-in fiber F**1** on a tip side aligned with (exposed at) the connection end surface **42** of the ferrule **40**. After the built-in fiber F**1** is inserted through the inside of the biasing member **60** and the second member **70**, the first member **50** and the second member **70** are connected. Specifically, the sliding tubular portion **52** is inserted inside the surrounding tubular portion **71** with the biasing member **60** set on an outer side of the surrounding tubular portion **71**. At this time, the slide rib **53** is positioned in the slide groove **72**, and the retainer **54** is positioned in the hole **75** (see FIG. **4**).

(39) When the built-in fiber F**1**, the ferrule **40**, the first member **50**, the biasing member **60**, and the second member **70** are combined in this manner, a part of the built-in fiber F**1** on the base end side becomes a state of extending rearward from the sleeve fixing portion **73** of the second member **70**. Note that, in this state, the base end side of the built-in fiber F**1** is coated with a resin for protection. Therefore, peeling off of the coating resin is generally performed before the fusion splicing.

(40) Next, the base end side of the built-in fiber F**1** from which the coating has been peeled off and the connection fiber F**2** are fusion-spliced. At this time, the connection fiber F**2** and the protection unit **80** are inserted through the inner housing **30** in advance.

(41) Next, the protection sleeve **81** is fixed to the sleeve fixing portion **73**. Specifically, the sleeve fixing portion **73** is covered with the protection sleeve **81**, and the protection sleeve **81** is heated. With the heating, the protection sleeve **81** is heat-shrunk to fasten the protrusions **73***a* of the sleeve fixing portion **73**. Also, the adhesive layer **84** disposed inside of the protection sleeve **81** softens and then hardens as it cools. Thereby, the protection sleeve **81** is adhesively fixed to the sleeve fixing portion **73**. At this time, since the tensile strength member **82** is sandwiched between the sleeve fixing portion **73** and the protection sleeve **81**, the tensile strength member **82** is also fixed to the sleeve fixing portion **73**.

(42) Next, the inner housing **30** is moved to the front side with respect to the holding part **2** (the second member **70**). Thereby, the state illustrated in FIG. **5** is obtained. Although not illustrated in FIG. **5**, when the key portion **76** of the holding part **2** enters the inside of the key groove **35** of the inner housing **30**, rotation of the holding part **2** with respect to the inner housing **30** is restricted. Also, the protection unit **80** is fixed to the holding part **2**. Therefore, it is suppressed that relative positions between the pair of tensile strength members **82** and the pair of tab portions **32** in the circumferential direction are deviated. In other words, it becomes a state in which the pair of tensile strength members **82** are positioned between the pair of tab portions **32**.

(43) Next, as illustrated in FIG. **7**, the outer housing **20** is attached to the inner housing **30**. More specifically, the ferrule **40**, the holding part **2**, and the inner housing **30** are inserted into the outer housing **20** from the rear side of the outer housing **20**. At this time, since the inclined surface **36***a* and the inclined surface **33***b* are formed on the protruding portion **36** and the locking protrusion **33***a* of the inner housing **30**, insertion work is facilitated. The outer housing **20** and the inner housing **30** are connected when the locking protrusion **33***a* of the inner housing **30** enters the inside of the locking hole **21** of the outer housing **20**.

(44) Here, when the inner housing **30** is inserted into the outer housing **20**, the protruding portion **36** is pushed inward in the radial direction by an inner wall of the outer housing **20**. This is because a distance between outer ends of a pair of protruding portions **36** in the radial direction before the inner housing **30** is inserted into the outer housing **20** is larger than a dimension of an inner space of the outer housing **20** in the first direction Z. When the protruding portion **36** is pushed, the tab portion **32** is tilted inward in the radial direction to press the protection sleeve **81**.

(45) When the tab portion **32** presses the protection sleeve **81**, the protection sleeve **81** is sandwiched between the tab portions **32** and the protrusions **73***a*. With this configuration, it is possible to increase a fixing force of the protection sleeve **81** with respect to the sleeve fixing

portion **73**. It is possible to represent the fixing force of the protection sleeve **81** with respect to the sleeve fixing portion **73** by, for example, a magnitude of a force when the protection sleeve **81** is pulled backward and the protection sleeve **81** comes out of the sleeve fixing portion **73**.

(46) Also, each of the pair of tensile strength members **82** is positioned between the tab portions **32** and the sleeve fixing portion **73**. Accordingly, the tensile strength members **82**, together with the protection sleeve **81**, are also sandwiched between the tab portions **32** and the protrusions **73***a*. It is possible to more firmly fix the protection sleeve **81** and the tensile strength members **82** to the sleeve fixing portion **73**, compared to a case in which the protection sleeve **81** and the tensile strength members **82** are fixed to the sleeve fixing portion **73** only by a heat shrinkage force of the protection sleeve **81**.

(47) With the tab portion **32** tilted inside the outer housing **20**, the contact surface **36***b* of the protruding portion **36** is in contact with the inner surface of the outer housing **20**. Here, as illustrated in FIG. **7**, the contact surface **36***b* is inclined with respect to the inner surface of the outer housing **20**. Therefore, only a rear end portion of the contact surface **36***b* is in contact with the outer housing **20**. According to the present configuration, compared to a case in which the entire contact surface **36***b* is brought into contact with the outer housing **20**, it is possible to more stabilize a tilt amount of the tab portion **32** inward in the radial direction. As the tilt amount of the tab portion **32** is stabilized, a pressure applied on the protection sleeve **81** is also stabilized.

(48) Thereafter, the assembly of the optical connector **1** is completed by attaching the case **10** to an outer side of the outer housing **20**.

(49) As described above, the optical connector **1** of one or more embodiments includes the ferrule **40** holding the optical fiber F, the holding part **2** holding the ferrule **40**, the protection sleeve **81** whose tip is fixed to the sleeve fixing portion **73** of the holding part **2**, the inner housing **30** inside which at least a part of the sleeve fixing portion **73** is housed, and the outer housing **20** inside which at least a part of the inner housing **30** is housed, in which the inner housing **30** includes a tab portion **32** which is able to be tilted in a radial direction, and the tab portion **32** is configured to apply a pressure on the protection sleeve **81** by being tilted inward in the radial direction in a state in which the tab portion **32** is positioned inside the outer housing **20**. According to the optical connector **1** having such a configuration, by positioning the tab portion **32** inside the outer housing **20**, it is possible to tilt the tab portion **32** inward in the radial direction to apply a pressure on the protection sleeve **81**. Due to the pressure, it is possible to fix the protection sleeve **81** to the sleeve fixing portion **73**. Therefore, it is possible to stabilize a fixed state, compared to a case in which the protection sleeve **81** is fixed only by a structure whose fixed state is affected by heating such as, for example, a heat shrinkage force of the protection sleeve **81** or adhesion by the adhesive layer **84**.

(50) Also, as illustrated in FIG. **7**, the tab portion **32** is disposed at a position facing the sleeve fixing portion **73** in the first direction Z. According to this configuration, it is possible to clamp the protection sleeve **81** between the tab portion **32** and the sleeve fixing portion **73** by using the force of the tab portion **32** pressing the protection sleeve **81** inward in the radial direction. Therefore, it is possible to further increase the fixing force of the protection sleeve **81** with respect to the sleeve fixing portion **73**.

(51) Also, the sleeve fixing portion **73** includes the plurality of protrusions **73***a* disposed at intervals in the longitudinal direction X and protruding outward in the radial direction, and the tab portion **32** is disposed at a position facing at least a part of the plurality of protrusions **73***a* in the first direction Z. According to this configuration, it is possible to make the protrusion **73***a* bite into the protection sleeve **81** by the pressure of the tab portion **32** applied on the protection sleeve **81**. Therefore, it is possible to further increase the fixing force of the protection sleeve **81** with respect to the sleeve fixing portion **73**.

(52) Also, as illustrated in FIG. **6**, the inner circumferential surface **32***a* of the tab portion **32** has an arcuate shape when viewed from the longitudinal direction X. According to this configuration, it is possible to increase a contact area between the tab portion **32** and the protection sleeve **81** when the

tab portion **32** is tilted inward in the radial direction. Therefore, it is possible to apply the pressure on the protection sleeve **81** more stably.
(53) Also, the tab portion **32** has the contact surface **36***b* that comes into contact with the outer housing **20** in a state in which the pressure is applied on the protection sleeve **81**. As illustrated in FIG. **7**, the contact surface **36***b* is inclined with respect to the inner surface of the outer housing **20**. According to this configuration, compared to a case in which the entire contact surface **36***b* comes in contact with the inner surface of the outer housing **20**, a tilt amount of the tab portion **32** directed inward in the radial direction is stabilized. Therefore, it is possible to stabilize the pressure of the tab portion **32** applied on the protection sleeve **81**.
(54) Also, the tensile strength member **82** is disposed inside of the protection sleeve **81**, and the tensile strength member **82** is positioned between the tab portion **32** and the sleeve fixing portion **73**. According to this configuration, it is possible to fix the tensile strength member **82** to the sleeve fixing portion **73** by using the pressure of the tab portion **32** applied on the protection sleeve **81**.
(55) The technical scope of the present invention is not limited to the above-described embodiments, and it is possible to make various modifications in a range not departing from the meaning of the present invention.
(56) For example, in the above-described embodiments, the inner housing **30** includes two tab portions **32**. However, it is possible to change the number of the tab portions **32** of the inner housing **30**. The number of the tab portions **32** may be one, or may be three or more.
(57) Also, it is possible to change the number of the tensile strength members **82** disposed inside of the protection sleeve **81**. The number of the tensile strength members **82** may be one, or three or more.
(58) Also, in the above-described embodiments, the tab portion **32** and the sleeve fixing portion **73** are disposed to face each other in the radial direction. However, the tab portion **32** may not necessarily have to face the sleeve fixing portion **73**. For example, even when the tab portion **32** is positioned behind the sleeve fixing portion **73**, it is possible to make the protection sleeve **81** bite into the protrusions **73***a* of the sleeve fixing portion **73** by the tab portion **32** applying a pressure on the protection sleeve **81** inward in the radial direction. Similarly, the tab portion **32** may not be disposed at a position facing the protrusions **73***a*.
(59) Although the disclosure has been described with respect to only a limited number of embodiments, those skilled in the art, having benefit of this disclosure, will appreciate that various other embodiments may be devised without departing from the scope of the present invention. Accordingly, the scope of the invention should be limited only by the attached claims.
REFERENCE SIGNS LIST
(60) **1** Optical connector **2** Holding part **20** Outer housing **30** Inner housing **32** Tab portion **32***a* Inner circumferential surface **36***b* Contact surface **40** Ferrule **73** Sleeve fixing portion **73***a* Protrusion **81** Protection sleeve **82** Tensile strength member F Optical fiber X Longitudinal direction

Claims

1. An optical connector comprising: a ferrule holding an optical fiber; a holder holding the ferrule; a protection sleeve comprising a tip that is fixed to a sleeve fixing portion of the holder; an inner housing in which at least a part of the sleeve fixing portion is housed; and an outer housing in which at least a part of the inner housing is housed, wherein the inner housing comprises a tab portion that: tilts in a radial direction of the ferrule, and applies a pressure on the protection sleeve by tilting inward in the radial direction in a state in which the tab portion is disposed inside the outer housing.
2. The optical connector according to claim 1, wherein the tab portion is disposed at a position facing the sleeve fixing portion.

3. The optical connector according to claim 2, wherein the sleeve fixing portion comprises protrusions: disposed at intervals in a longitudinal direction of the ferrule, and protruding outward in the radial direction, and the tab portion is disposed at a position facing at least a part of the protrusions.
4. The optical connector according to claim 1, wherein an inner circumferential surface of the tab portion has an arcuate shape when viewed from a longitudinal direction of the ferrule.
5. The optical connector according to claim 1, wherein the tab portion has a contact surface that comes in contact with the outer housing in a state in which a pressure is applied on the protection sleeve, and the contact surface is inclined with respect to an inner surface of the outer housing.
6. The optical connector according to claim 1, wherein a tensile strength member is disposed inside of the protection sleeve and between the tab portion and the sleeve fixing portion.